

Introduction

Digital Twins facilitate robot data collection and training with minimal sim-to-real gap. Yet, importing a realistic scene into simulation is a costly job. Thus, We present an effective method to reconstruct real-world scenes with high fidelity.

Research Goal

1. Facilitate real-to-sim process
2. Automate object articulation
3. Enhance manipulation and data collection process

Method Overview

A. Initial Reconstruction

1. Generate initial 3D Gaussian Splats and meshes through the GS2Mesh Pipeline.¹
2. Separate object GS from background using SAM masks and voting-based filtering.



B. Refinement

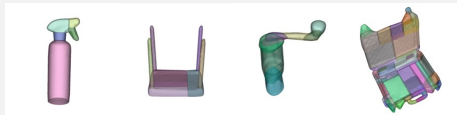
Apply Mesh Processing

Filtering

- | | |
|---------------------------|--------------------------|
| 1. Component Cleaning | 5. Isotropic Remeshing |
| 2. Surface Reconstruction | 6. Occlusion Repair |
| 3. Artifact Removal | 7. Hole Closing |
| 4. Simplification | 8. Attribute Restoration |

C. Mesh Decomposition

Decompose mesh into non-overlapping fragments using ACD.² (assuming extended links)



D. Mesh Segmentation

1. Group fragments to define each link.
2. Map vertices based on SDF/proximity and bind GS using KNN voting.



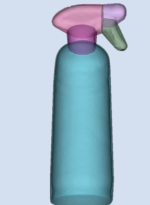
Real-to-Sim Process



Refined Mesh



Convex Decomp.



Segmented



E. Articulation

Detect Boundary Loops (for each part)
Detect loops through DFS traversal and greedy selection with uniqueness.

Estimate Separating Planes (for each loop)
Estimate plane axes on k-hop local geometry.

- Single Plane: Linear SVM
- Multi-plane: EM Algorithm (and/or criteria)

Fuse Reciprocal Loop Pairs (for each interface)
Interfaces are detected twice per connection. Thus, match and fuse loop into Reciprocal Loop Pairs using Hungarian Algorithm, minimizing geometric cost. (proximity + normal alignment)

Select Joints (for each connection)

- Suggest joint candidates (plane normal, intersection, or multi-boundary axis).
- Adjust and finalize joint kinematics.

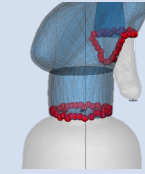


F. Generate Articulated Object

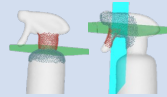
Export a holistic URDF and synchronized Gaussian Splat files per parts.



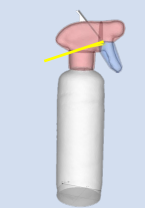
Boundary Loops



Plane Estimation



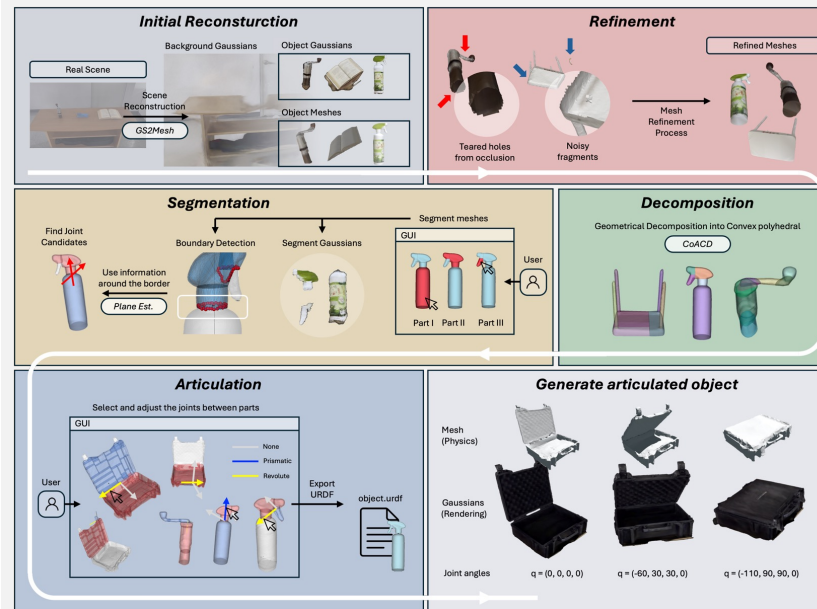
Joint Candidates



Articulated Pair



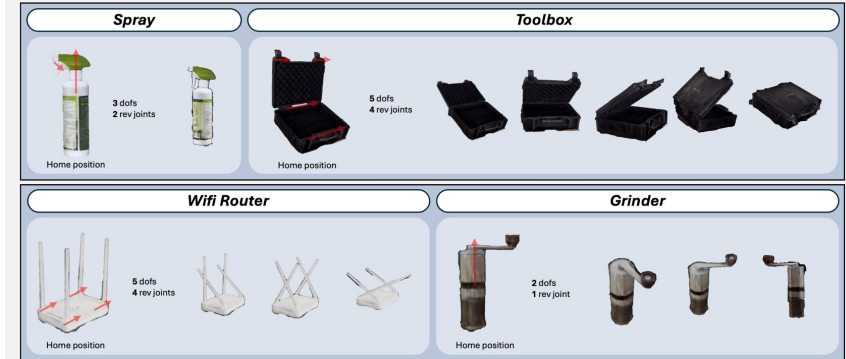
Overall Pipeline diagram



Experiments

Reconstruction Results

- Successfully generated photorealistic 3DGS-Mesh Pairs having realistic collision and kinematics.
- Validated pipeline on diverse real-world objects.



Manipulation Demo / IL Training

Performed dexterous tasks with an Inspire Hand in the reconstructed environment. Implemented a data collection system and trained a Diffusion Policy for grasping task.

- **Teleoperation Device:** VIST³
- **Simulator:** HXR Simulator⁴ (based on UE 5.3)

Box Opening



Spraying



Grinding



Conclusion

- Proposed a semi-automated pipeline to reconstruct articulated objects from a single video.
- Achieved both highly realistic rendering and physical interaction, reducing reconstruction cost.
- Demonstrated system efficacy through simulated teleoperation and IL training for manipulation tasks.
- As future works, we will (1) verify zero-shot sim2real transfer performance, (2) integrate semantic segmentation method, (3) conduct quantitative comparison against VLM based articulation tools.

References

- [1] Y. Wolf et al., GS2Mesh: Surface Reconstruction from Gaussian Splatting via Novel Stereo Views, ECCV, 2024.
- [2] X. Wei et al., Approximate Convex Decomposition for 3D Meshes with Collision-Aware Concavity and Tree Search, ACM TOG, 2022.
- [3] Y. Lee et al., Visual-Inertial Hand Motion Tracking with Robustness against Occlusion, Interference, and Contact, Science Robotics, 2021.
- [4] H. Ji et al., GPU-Accelerated Subsystem-Based ADMM for Large-Scale Interactive Simulation, ICRA, 2025.